

A photograph of a modern university building with large glass windows and a white facade, situated behind a field of green grass with many yellow dandelions. The sky is blue with a few white clouds. A semi-transparent white box is overlaid on the left side of the image, containing the course title and tutorial information.

5CC512 Communication and Security Protocols

Week 01 Tutorial Mathematical Preliminaries

Agenda

- Power of a Number
- Binary Number System
- Logical Operators
- Modular Arithmetic
- Random Numbers
- Prime Numbers

Power of a Number

- Also known “exponent of a number”
- It says how many times to use the number in a multiplication.
 - It is written as a small number to the right and above the base number.
- $5^2 = 5 \times 5 = 25$
 - 5 squared
- $5^3 = 5 \times 5 \times 5 = 125$
 - 5 cubed
- $5^4 = 5 \times 5 \times 5 \times 5 = 625$
- $5^5 = 5 \times 5 \times 5 \times 5 \times 5 = 3125$
- $5^0 = 1$
 - $(\text{any number})^0 = 1$



Exercise

- Please calculate the following numbers

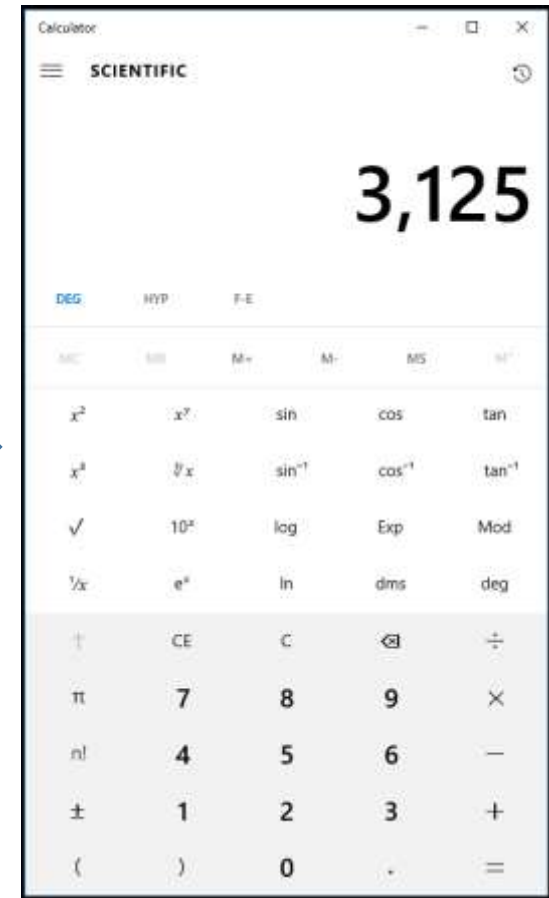
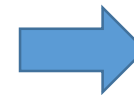
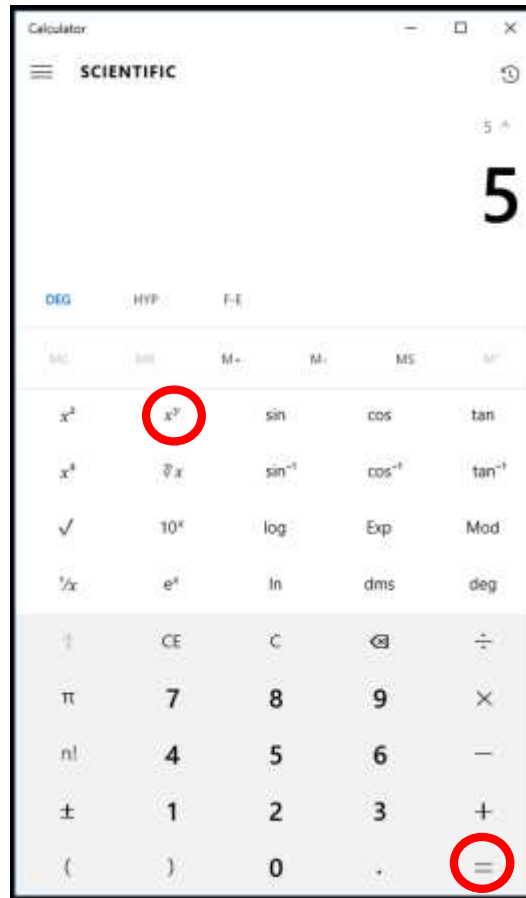
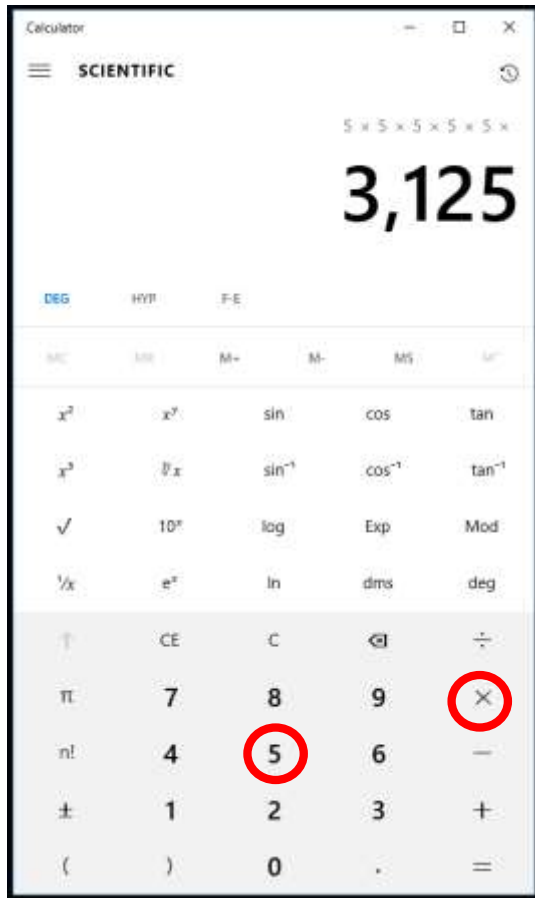
- $4^3 = \underline{\hspace{2cm}}$ $4^3 = 4 \times 4 \times 4 = 64$

- $2^5 = \underline{\hspace{2cm}}$ $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

- $6^4 = \underline{\hspace{2cm}}$ $6^4 = 6 \times 6 \times 6 \times 6 = 1296$

Power of a Number

- Using Calculator



Exercise

- Calculate 123^{12} using a calculator?
- Very difficult to calculate by hand
 - $123 \times 123 \times 123 \times 123 \times 123 \times 123 \times 123 \times 123 \times 123 \times 123 \times 123 = ???$



Any Questions?



Binary Number System

- Why?

- Computers use the binary number system to manipulate and store all of their data including numbers, words, videos, graphics, and music.

- Also called base-2 system

- representing numbers that counts by using combinations of only two numerals: zero (0) and one (1).

- Reminder : base-10 system

- We use in daily life.
- Ten numerals : 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- $4 + 5 = 9$

- $9 + 4 = 13 = 10 + 3$

- $4572 = 4 \times 1000 + 5 \times 100 + 7 \times 10 + 2 \times 1$

- $= 4 \times 10^3 + 5 \times 10^2 + 7 \times 10^1 + 2 \times 10^0$

Number → 4 5 7 2

Digit → 3 2 1 0

Binary Number System

- Base-2

- $1 + 0 = 1, 0 + 1 = 0$

- $1 + 1 = \text{X}$

- $1 + 1 = 10$

- $= 1 \times 2 + 0 \times 1 = 2 + 0$

- $1011 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

- $= 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$

- $= 8 + 0 + 2 + 1 = 11$

Number $\rightarrow$ 1 0 1 1

Digit $\rightarrow$ 3 2 1 0

1010100111101101

Binary Number System

- Adding two numbers

- Base-10

$$\begin{array}{r} 5 \\ 7 \\ + \text{-----} \\ 12 \end{array}$$

- Base-2

$$\begin{array}{r} 0101 \\ 0111 \\ + \text{-----} \\ 1100 \end{array}$$

Binary Number System

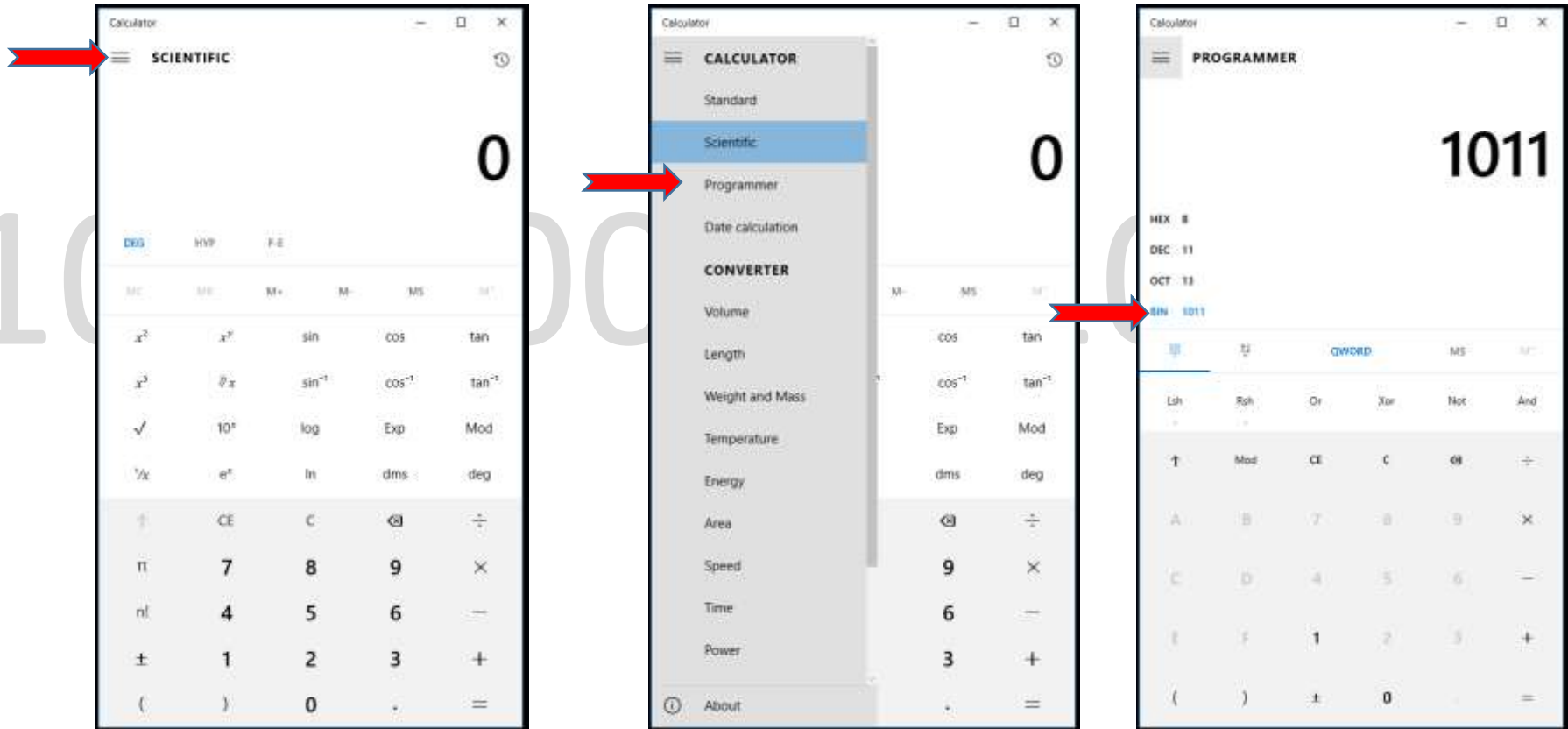
- Calculate the following

- $$\begin{array}{r} 1000 \\ 0111 \\ + \text{-----} \\ 1111 \\ 1010 \\ 1001 \\ + \text{-----} \\ 1\ 0011 \end{array}$$

1010100111101101

Binary Number System

- TIP : Use a Programmer Calculator



Binary Number System

- Typical number of digits
 - 8 bit (byte) : 1010 1111
 - 16 bit : 1001 1110 1111 0011
 - 32 bit : 1001 1110 1111 0011 1101 1010 1001 1011
 - 64 bit :
 - 1001 1110 1111 0011 1101 1010 1001 1011 1001 1110 1111 0011 1101 1010 1001 1011

- We use Base-16 (Hex) to make it better looking

- Numerals?
 - 0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15

- 15 : is it fifteen or $15 = 1 \times 16 + 5 = 21$ (twenty one)?



- 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

- $15 \rightarrow F$, $21 \rightarrow 15$

Binary Number System

- Use calculator to find Binary and Hex equivalents of the following Base-10 (decimal) numbers:

• 45	=	<u>0010 1101</u>	<u>0000 002D</u>
• 128	=	<u>1000 0000</u>	<u>0000 0080</u>
• 2391	=	<u>1001 0101 0111</u>	<u>0000 0957</u>

- Use calculator to find Binary and Decimal equivalents of the following Base-16 numbers:

• C5	=	<u>1100 0101</u>	<u>197</u>
• 10A	=	<u>0001 0000 1010</u>	<u>266</u>
• 3BF2	=	<u>0011 1011 1111 0010</u>	<u>15,346</u>



Connection between Binary and Hex Number Systems

- C5 = 1100 0101
- 10A = 0001 0000 1010
- 3BF2 = 0011 1011 1111 0010



1010100111101101

C 5
↓ ↓
1100 0101

3 B F 2
↓ ↓ ↓ ↓
0011 1011 1111 0010

- Each Hex digit corresponds to **four** binary digits.

Any Questions?



Logical operators – Boolean Operators

- Used to evaluate data in computers
 - Not, And, Or, Exclusive-Or (XOR)
- Truth Tables :

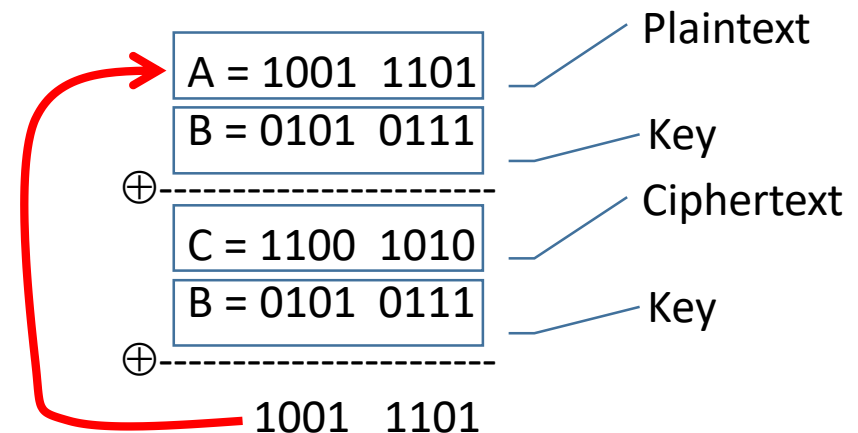
NOT	
x	x'
0	1
1	0

AND		
x	y	xy
0	0	0
0	1	0
1	0	0
1	1	1

OR		
x	y	x+y
0	0	0
0	1	1
1	0	1
1	1	1

XOR		
x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

- Why is XOR useful in Cyber Security?
 - Promotes inverse operation very easily



Logical operators – Boolean Operators

- Please calculate

- $$\begin{array}{r} 1101\ 0001 \\ 1001\ 0110 \\ \oplus \text{-----} \\ 0100\ 0111 \end{array}$$

- $$\begin{array}{r} 1101\ 0001 \\ 1001\ 0110 \\ \text{AND} \text{-----} \\ 1001\ 0000 \end{array}$$

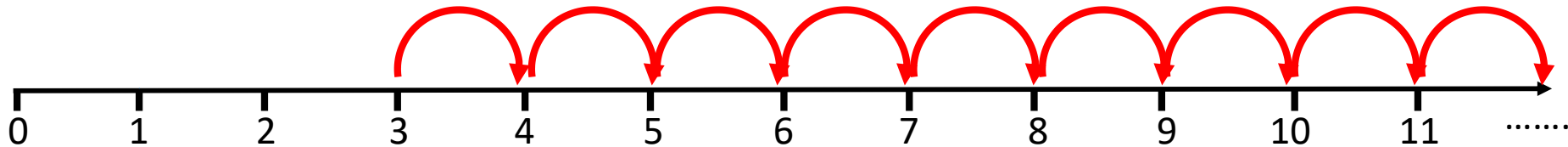
- $$\begin{array}{r} 1101\ 0001 \\ 1001\ 0110 \\ \text{OR} \text{-----} \\ 1101\ 0111 \end{array}$$

Any Questions?



Modular Arithmetic

- What is modular arithmetic?
 - A mathematical system of arithmetic integers.
 - Numbers **wrap around** when reaching a certain value.

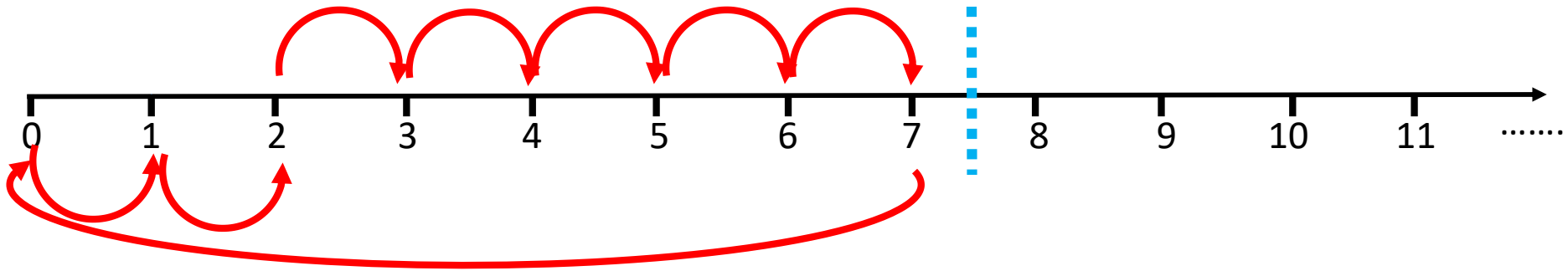


- Consider only adding numbers

$$3 + 5 = 8 + 10 = 18 + 120 = 138 + 2530 = 2668 + \dots + \dots + \dots \rightarrow \infty$$

Modular Arithmetic

- What is modular arithmetic?
 - We want to numbers **wrap around** when reaching a certain value.
 - This value is named as **modulus**.
 - Let's this value 8



$2 + 3 = 5$ + $5 = 10$ **X** not allowed, we bumped the barrier

For the last step, we go back to 0. **Thus $5 + 5 = 2$!!!!**

To differentiate this from normal arithmetic, we write **$5 + 5 = 2 \pmod{8}$**

It can be write as **$5 + 5 \equiv 2 \pmod{8}$** or **$5 + 5 \cong 2 \pmod{8}$** . We say “10 is congruent to 2 modulo 8”

Please note that all numbers that can be achieved in this system is between **0 and 7**.

↓
(8 - 1)

Modular Arithmetic

- Can you give any example in which we can use modular arithmetic?

- CLOCK !!!

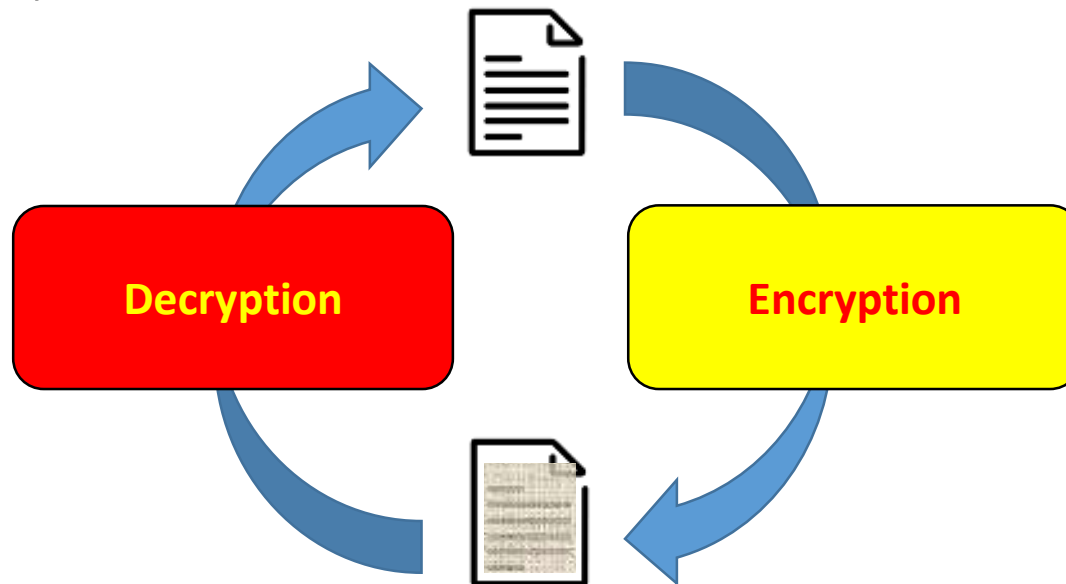


- After 12.00 it wraps back to 1
 - Actually it is between 0 and 11, but we use 12 instead of 0 for convenience.

$$7\text{am} + 3 = 10\text{am} + 4 = 14 \equiv 2\text{pm}$$

Modular Arithmetic

- Why modular arithmetic is important in Security?
- How do our cipher systems work?



- We will use this concept in
 - Diffie-Hellman Key Exchange
 - RSA Algorithm
 - Generating Pseudo Random Numbers

Modular Arithmetic

- Practical Approach
- How we can find out the answer for

$$345 + 568 \equiv ? \pmod{234}$$

or

$$12345 \equiv ? \pmod{36}$$

- Counting is not practical in the first case.
- We do not know how 12345 is obtained in the second case.
 - You can guess any two numbers added up to 12345
 - 10000 + 2345 or 12000 + 345 or 8967 + 3378
 - But still not practical!!!!

Modular Arithmetic

- Method 1 :
- Subtract the **modulus** from the total number successively until to reach a result less than the **modulus**
- $345 + 568 \equiv ? \pmod{234}$

$$345 + 568 = 913$$


$$913 - 234 = 679 - 234 = 445 - 234 = 211$$


$$345 + 568 \equiv 211 \pmod{234}$$

Modular Arithmetic

- What number does 108 congruent to in modulo 15? Show your work out using subtraction method

$$108 - 15 = 93 - 15 = 78 - 15 = 63 - 15 = 48 - 15 = 33 - 15 = 18 - 15 = \mathbf{3}$$

Modular Arithmetic

- Method 1 :
- Subtract the **modulus** from the total number successively until to reach a result less than the **modulus**
- Still cumbersome!!
 - How many subtractions do we need for $12345 \equiv ? \pmod{36}$
 - Can you tell me?

- 342



Modular Arithmetic

- Method 2 :

- Use Integer Division
- Divide the total sum by the **modulus**
- The remainder is the answer !!!
- $345 + 568 \equiv ? \pmod{234}$

$$345 + 568 = 913$$

$$\text{Normal division : } 913 / 234 = 3.901709\ldots$$

$$\text{Integer division : } 913 / 234 = 3 \text{ with remainder } 211$$

 quotient

- How can we find the remainder?

$$\text{Normal division : } 913 / 234 = 3.901709\ldots$$

$$913 - 234 \times 3 = 913 - 234 \times 3 = 913 - 702 = 211$$

Modular Arithmetic

- Method 2 :

- Use Integer Division
- Divide the total sum by the modulus
- The remainder is the result
- 345

This shows how I can quickly calculate how many subtractions we do need
for $12345 \equiv ? \pmod{36}$

$$12345 / 36 = 342.916666....$$

$$12345 - 342 \times 36 = 12345 - 12312 = 33$$

$$\underline{12345 \equiv 33 \pmod{36}}$$

- How

N

$$913 - 234 \times 3 = 913 - 234 \times 3 = 913 - 702 = \mathbf{211}$$

Modular Arithmetic

- What number does 108 congruent to in modulo 15? Show your work out using division method

$$108 / 15 = 7.2$$

$$108 - 7 \times 15 = 108 - 105 = 3$$

Any Questions?

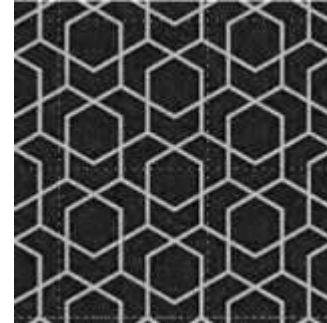
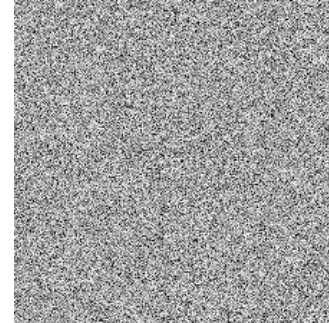


Random Numbers

- Why do we need random numbers in Cyber Security?

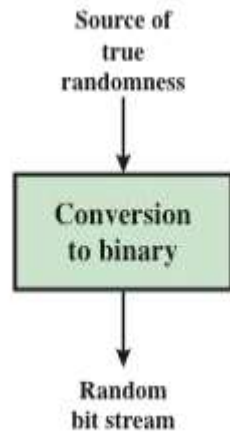
Random Numbers

- Why do we need random numbers in Cyber Security?
- Randomness : occurring without a pattern
- Unpredictability : cannot predict what is the next based on the previous numbers in the sequence
 - Shuffling characters in a text
 - Generating well-mixed passwords
 - Key generation
 - Nonces
 - Number Once
 - Used for handshaking to prevent replay attacks
- The opponent cannot guess the actual work

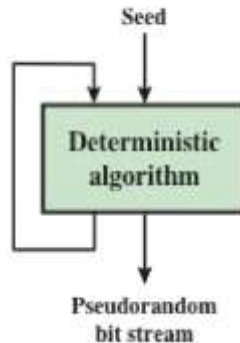


Random Numbers

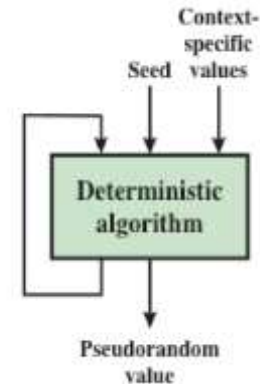
- Random and Pseudo Random Generators



True random number generator
(TRNG)



Pseudo random number generator
(PRNG)



Pseudo random Function
(PRF)

- Which one is better to use in Security Applications ?
- **Pseudo Random Number Generator**
 - Generating pure random numbers are difficult in classical computers
 - In cryptographic algorithms the sender and receiver should follow the same sequence
 - Cannot be achieved with True Random Numbers and PRF

Random Numbers

- Using modular arithmetic, a computer can generate numbers that appear random

- **Linear congruent generators**

- Start from a predetermined number (seed) : s
 - Secret between the sender and the receiver
- Based on two predetermined numbers : a (multiplier), m (modulus)
 - Algorithm specific
 - Available
- Generate sequence with modular arithmetic

$$r_0 = s$$

$$r_1 = s \times a^1 \pmod{m}$$

$$r_2 = s \times a^2 \pmod{m}$$

$$r_3 = s \times a^3 \pmod{m}$$

$$r_4 = s \times a^4 \pmod{m}$$

.....



$$r_0 = s$$

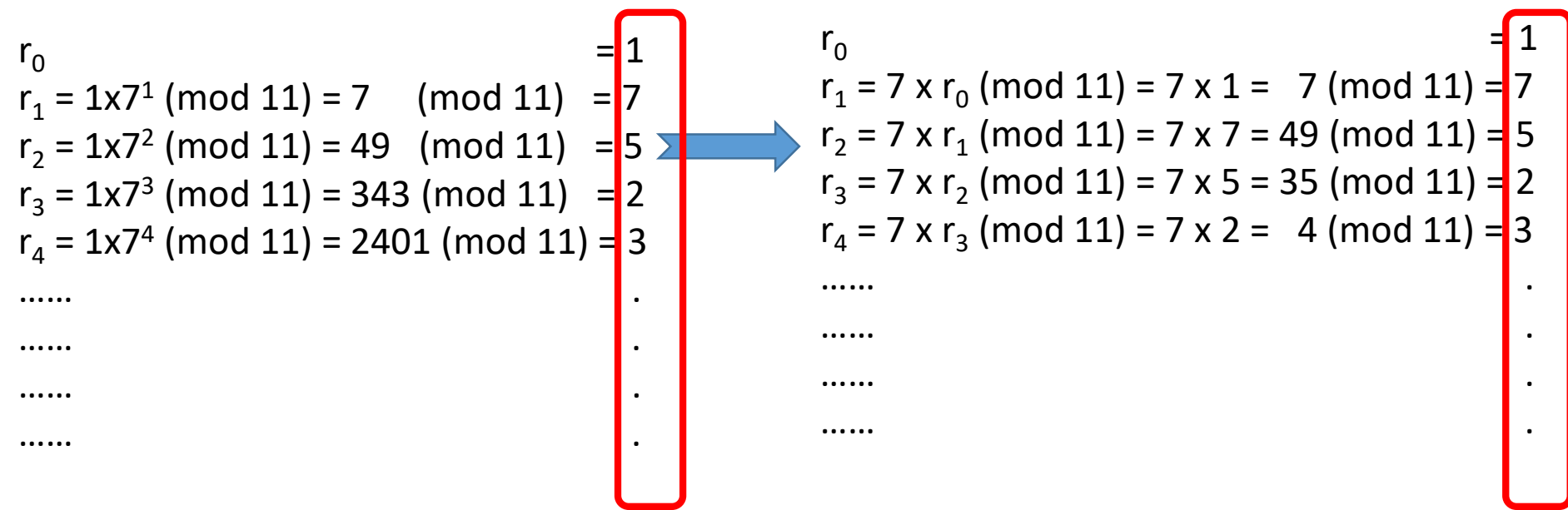
$$r_i = a \times r_{i-1} \pmod{m}$$

- Randomness depends on multiplier (a) and modulus (m)
 - Determining good choices of a and m is a research problem

Random Numbers

• Example :

• $s = 1, a = 7, m = 11$



$r = 1, 7, 5, 2, 3, \dots$

Random Numbers

- $s = 1, a = 7, m = 11$

$$\begin{aligned} r_0 &= 1 \\ r_1 &= 1 \times 7^1 \pmod{11} = 7 \pmod{11} = 7 \\ r_2 &= 1 \times 7^2 \pmod{11} = 49 \pmod{11} = 5 \\ r_3 &= 1 \times 7^3 \pmod{11} = 343 \pmod{11} = 2 \\ r_4 &= 1 \times 7^4 \pmod{11} = 2401 \pmod{11} = 3 \\ &\dots \\ &\dots \\ &\dots \\ &\dots \end{aligned}$$



$$\begin{aligned} r_0 &= s \\ r_i &= a \times r_{i-1} \pmod{m} \end{aligned}$$

$$\begin{aligned} r_0 &= 1 \\ r_1 &= 7 \times r_0 \pmod{11} = 7 \times 1 = 7 \pmod{11} = 7 \\ r_2 &= 7 \times r_1 \pmod{11} = 7 \times 7 = 49 \pmod{11} = 5 \\ r_3 &= 7 \times r_2 \pmod{11} = 7 \times 5 = 35 \pmod{11} = 2 \\ r_4 &= 7 \times r_3 \pmod{11} = 7 \times 2 = 14 \pmod{11} = 3 \\ &\dots \\ &\dots \\ &\dots \\ &\dots \end{aligned}$$

- Calculate 5th element in the sequence using both methods

$$r_5 = 1 \times 7^5 \pmod{11} = 16807 \pmod{11} = \mathbf{10}$$

$$r_5 = 7 \times r_4 \pmod{11} = 7 \times 3 = 21 \pmod{11} = \mathbf{10}$$

Random Numbers

- Other Pseudo Random Number Generator (PGNR)
 - Yarrow Algorithm
 - Naor–Reingold pseudorandom function
 - Blum Blum Shub (B.B.S.)
 - Squares RNG
 - Xoroshiro128+
 -
- List of PGNR
 - https://en.wikipedia.org/wiki/List_of_random_number_generators
- Randomness should be tested before using the algorithm
 - 2,3,7,4,6,8,1,2,3,7,4,6,8,1,2,3,7,....

period
 - We need a long period for better security.
 - Testing the randomness is another research area

Any Questions?



Prime Numbers

- A natural number
- Greater than 1
- **NOT** a product of two smaller numbers
 - A prime number is a positive whole number that is only divisible by **1** and **itself**.

Is 1 a prime number ?

NO! Why?

The number 1 is divisible by 1, and it is divisible by itself.
But itself and 1 are not two distinct factors.

An interesting story of 1.

- is it a number?
- is it even or odd?

Is 2 a prime number ?

Yes - $2 = 2 \times 1$

Prime Numbers

Is 17 a prime number?

YES – $17 = 1 \times 17$

Is 64 is a prime number?

NO – $64 = 1 \times 64 = 2 \times 32 = 4 \times 16 = 8 \times 8$

- Some prime numbers:

- 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, ..

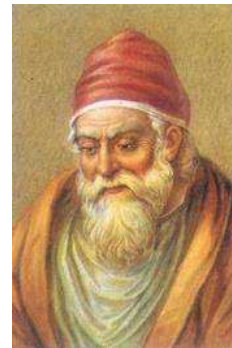
What is the largest prime number?

There is no largest prime number.

There are infinite number of prime numbers.

Euclid's theorem

https://en.wikipedia.org/wiki/Euclid's_theorem



Prime Numbers

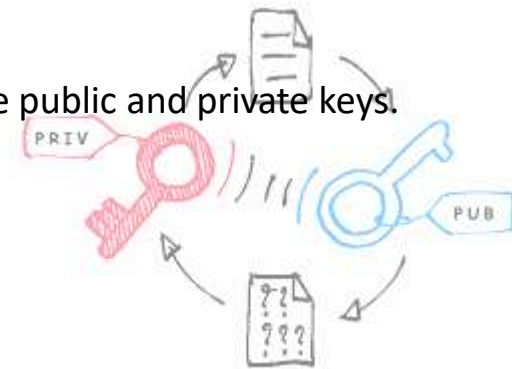
- Finding larger prime numbers is a holy grail for mathematicians and computer scientists
- The largest known prime number
 - $2^{82,589,933} - 1$
 - Guess how many digits has it got?
 - 24,862,048 digits
 - Homework : how many pages required to write down this number using MS Word in a standard page and a font size of 11 ?
- Prizes
 - Check : https://en.wikipedia.org/wiki/Largest_known_prime_number



Prime Numbers

Why are primes number important for us?

- They are fundamental to public key cryptography.
 - RSA uses a number which is a product of two prime numbers to calculate public and private keys.
 - $N = P \times Q$
 - Public key and Private key is derived used N , P and Q .
 - **P and Q must be kept secret!**
 - **Public key consists of N .**
 - **Therefore it is open to everyone!**
- **P and Q should be large enough so that finding them factorising N is practically impossible.**



Any Questions?



Software we use in the coming weeks

- Visual Studio Community
- <https://visualstudio.microsoft.com/vs/community/>



- Wireshark Network Analyzer
- <https://www.wireshark.org/>



- **You can DOWNLOAD and INSTALL them in your computer**